MILITARY SPECIFICATION

RESISTOR NETWORKS, FIXED, FILM,
ESTABLISHED RELIABILITY,
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-874, dated 21 June 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 1.2.1, delete and substitute the following PIN number:

" M874	H	01	A	1002	A	S
Military specification number (indicating MIL-R-874)	Characteristic (see 1.2.1.1)	Military specification number (indicating MIL-R-874/1) "	Schematic (see 1.2.1.4)	Resistance value (see 1.2.1.3)	Resistance tolerance (see 1.2.1.2)	Failure rate level (see 1.2.1.5)

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3.2, delete and substitute the following:

"3.2 Qualification. Networks furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.4 and 6.3). Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL."

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Add the following new paragraphs:

"3.5.2.1 Solder dip (retinning) leads. The manufacturer (or authorized category B or C distributor) may solder dip/retin the leads of product supplied to this specification, provided the solder dip process has been approved by the qualifying activity.

"3.5.2.2 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276 (NOTE: The 200 microinch maximum thickness is not applicable). The manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product.

b. When the lead originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.5.2.2a, approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the resistors are subjected to the dc resistance test (and other group A electricals). No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat test, followed by the moisture resistance test. (NOTE: Solder dip of gold-plated leads is not allowed.)

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"3.5.2.3 Solder dip/retinning options. The manufacturer (or authorized category B or C distributor) may solder dip/retin as follows:

a. After the 100 percent group A screening tests. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening tests shall be repeated on 100 percent of the lot. (NOTE: The manufacturer may solder dip/retin prior to the 100 percent electrical measurements of the group A, subgroup 1 tests.) The percentage defective allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests.

b. As a corrective action if the lot fails the group A solderability test.

c. After the group A inspection has been completed. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening test shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Following these tests, the manufacturer shall submit the lot to the group A solderability test as specified in 4.8.6."

Add the following new paragraph:

"3.5.5 Bonding. If thermocompression bonding is used for internal connections, a gold to gold bond shall form a solid phase weld (see 3.1)."

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* Add the following new paragraph:

"3.5.6 Tin plated finishes. Use of tin plating is prohibited as a final finish and as an undercoat effective 6 months from the date of this specification (see 6.4.3.1). Use of tin-lead (Sn-Pb) finishes are acceptable provided that the minimum lead content is 3 percent."

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TABLE VII, delete and substitute:

TABLE VII. Failure rate substitution.

Failure rate	Failure rate substitution
S	S
R	S, R
P	S, R, P
M	

TABLE VIII, delete and substitute:

TABLE VIII. Substitution of temperature characteristics.

Characteristic	Characteristic substitution
H	H
K	H, K
M	

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Table IX, delete and substitute:

* TABLE IX. Resistance tolerance substitution. *

Resistance tolerance	Resistance tolerance substitution
B	---
D	B
F	B, D
G	B, D, F
J	B, D, F, G

* Table X, delete and substitute:

* TABLE X. Qualification inspection. *

Inspection	Number of sample units	Requirement paragraph	Method paragraph	Number of 1/ failures allowed
Group I 2/ 3/				
Thermal shock	All sample units 4/	3.8	4.8.3	5/
Power conditioning		3.9	4.8.4	5/
DC resistance		3.10	4.8.5	5/
Group IA 3/ 6/				
Visual and mechanical inspection		3.1, 3.4, 3.5 3.28, 3.29	4.8.2	0
Group IB 7/				
Solderability		3.11	4.8.6	1
Group II 7/				
Resistance to solvents	12 4/	3.12	4.8.7	1
Group III				
Resistance-temperature characteristics	20 or 30	3.13	4.8.8	1 9/
Low temperature operation	10 high	3.14	4.8.9	1 9/
Short-time overload	10 critical 8/	3.15	4.8.10	1 9/
Terminal strength	10 low	3.16	4.8.11	1 9/
Group IV				
Thermal shock		3.8	4.8.3	1 9/
Dielectric withstanding voltage	20 or 30	3.17	4.8.12	1 9/
Insulation resistance	10 high	3.18	4.8.13	1 9/
Resistance to soldering heat 10/	10 critical 8/	3.19	4.8.14	1 9/
Moisture resistance	10 low	3.20	4.8.15	1 9/
Group V				
Shock (specified pulse)	20 or 30	3.21	4.8.16	1 9/
Vibration, high frequency	10 high	3.22	4.8.17	1 9/
	10 critical 8/			
	10 low			

See footnotes at end of table.

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" TABLE X. Qualification inspection - Continued.

Inspection	Number of sample units	Requirement paragraph	Method paragraph	Number of failures allowed
Group VI Life	20 or 30 10 high 10 critical 8/ 10 low	3.23	4.8.18	1 9/
Group VIA Power rating at 25°C	20 or 30 10 high 10 critical 8/ 10 low	3.23.1	4.8.18	1 9/
Group VII High temperature exposure Low temperature storage	20 or 30 10 high 10 critical 8/ 10 low	3.24 3.25	4.8.19 4.8.20	1 9/ 1 9/
Group VIII Fungus 4/ 11/	10	3.26	4.8.21	0

1/ Failure of a single network in one or more tests of a group shall be charged as a single defective.
2/ Tests need not be performed if group A inspection has been performed on the qualification samples.
3/ Nondestructive tests.
4/ Sample units for group II and VIII are not required to be subjected to group I.
5/ Networks shall meet all requirements of group I before subjecting to groups III through VII.
6/ Marking shall be considered defective if the marking is illegible or incorrect.
7/ Tests may be performed on electrical rejects.
8/ When no critical value is specified, only the highest and lowest resistance values shall be tested (20 samples, total).
9/ Number of defective packages allowed per resistance value tested (e.g., low, critical, and high resistance).
10/ The internal visual examination shall be performed after the final inspection of group IV.
11/ The fungus requirement is either by certification or performance."

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* 4.4.4.1c, delete and substitute the following:

"c. Procedure III - Maintenance of FR levels qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.7.2.1). For FR levels R and S data from all lead types may be combined."

4.5.1, delete in its entirety.

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titute the following table: (letters change in amendment 2/ * TABLE XII. Delete and subst

* 4.7.1.2.3, last sentence: Delete in its entirety.

After 4.7.1.2.3 and before table XI, add the following new paragraphs:

"4.7.1.2.4 Subgroup 4 (solderability).

"4.7.1.2.4.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. If there are one or more defects, the lot shall be considered to have failed.

"4.7.1.2.4.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.8.6. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in b.

* 4.7.1.2.4.2b delete and substitute the following paragraph (alters a change from amendment 1):

" b. The manufacturer submits the failed lot to a 100-percent solder dip using an approved solder dip process in accordance with 3.5.2.1. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Thirteen additional samples shall then be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot may be reworked a second time and retested. If the lot fails the second rework, the lot shall be considered rejected and shall not be furnished against the requirements of this specification."

"4.7.1.2.4.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract."

Table XI, add the following new subgroup:

"

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 4			
Solderability	3.11	4.8.6	See 4.7.1.2.4

"

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Table XII, delete and substitute the following table:

" TABLE XII. Sampling plans for ppa categories.

Lot size	Sample size
2 to 8	100 percent
9 to 15	100 percent
16 to 25	100 percent
26 to 50	100 percent
51 to 90	100 percent
91 to 150	125
151 to 280	192
281 to 500	192
501 to 1,200	192
1,201 to 3,200	192
3,201 to 10,000	192
10,001 to 35,000	294
35,001 to 150,000	294
150,001 to 500,000	345
500,000 and over	435

"

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Table XIII, delete in its entirety.

4.7.1.3, delete and substitute:

"4.7.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table XIV, in the order shown. Tests shall be performed on sample units that have been subjected to and have passed the group A inspection."

Table XIV, delete solderability and related entries from subgroup 2.

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* 4.8.5a, delete and substitute the following paragraph:

"a. Measuring apparatus: Different types of measuring test equipment (multimeters, bridges, or equivalent) are permitted to be used on the initial and final readings of this test, provided the equipment is the same style, model, or if it can be shown that the performance of the equipment is equivalent or better."

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* Table XVII, delete and substitute:

" TABLE XVII. Resistance-temperature characteristic.

Sequence	Temperature	
	Qualification inspection	Group B acceptance inspection 1/
	°C	°C
1	Room temperature 2/	Room temperature 2/
2	-15 ±3	-55 ±3
3	-55 ±3	Room temperature 2/
4	Room temperature 2/	125 ±3
5	65 ±3	---
6	125 ±3	---

1/ At the option of the manufacturer, the reverse sequence may be as follows:

1	Room temperature 1/
2	+125 ±3
3	Room temperature 1/
4	-55 ±3

2/ This temperature shall be considered the reference temperature for each of the succeeding temperatures. "

* Add the following new paragraph:

"6.4.3.1 Tin plating finishes. Tin plating is prohibited (see 3.5.6) since it may result in tin whisker growth. Tin whisker growth could adversely affect the operation of electronic equipment systems. For additional information on this matter, refer to ASTM B545 (Standard Specification for Electrodeposited Coating of Tin)."

30.3, delete and substitute:

"30.3 Description of items. The manufacturer shall submit a detailed description of the network being submitted for inspection, including materials used for the resistance element and protective enclosure or coating. After qualification has been granted, no changes shall be made in materials, design, or construction without proper notification to the qualifying activity."

40.1, delete and substitute:

"40.1 Extension of qualification. The resistance range included in the qualification of any one network style will be between the lowest and highest resistance values which pass the qualification inspection (see 3.1). Qualification of one characteristic is basis for qualification of another characteristic, as indicated in table XIX. Qualification of the lower resistance tolerances will qualify for the higher resistance tolerances in accordance with table XX. Extent of qualification by style shall be in accordance with table XXI. Internal welded construction will qualify a nonwelded construction. Extent of qualification by schematic shall be in accordance with table XXII. As a requisite for extension of qualification as described herein between characteristics, tolerances, schematics, resistance values and failure rates, the product involved must be manufactured using the same facilities, processes, and materials as the product originally submitted for qualification."

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Table XVIII, delete and substitute:

" TABLE XVIII. Critical resistance value for qualification inspection. 1/

Style	Critical resistance value 1/ 2/					
	Schematic A	Schematic B	Schematic C	Schematic G	Schematic J	Schematic H
				Characteristics		
	H, K, M	H, K, M	H, K, M	H, K, M	H, K, M	H, K, M
	MΩ	MΩ	MΩ	MΩ	MΩ	MΩ
RZR10	0.047	0.1			.200	
RZR11	0.047	0.1			.200	
RZR20	0.047	0.1			.200	
RZR21	0.047	0.1			.200	
RZR30	0.047	0.1			.160	
RZR31	0.047	0.1			.160	
RZR40			.012	.012		.020
RZR41			.012	.012		.020
RZR50			.012	.012		.020
RZR51			.012	.012		.020
RZR60			.012	.012		.020
RZR61			.012	.012		.020
RZR70			.020	.020		.033
RZR71			.020	.020		.033
RZR80			.020	.020		.033
RZR81			.020	.020		.033
RZR90			.020	.020		.033
RZR91			.020	.020		.033
RZR95	0.047	0.10			.160	
RZR96	0.047	0.10			.160	

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when maximum continuous working voltage is applied."

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Table XXI, delete and substitute:

" TABLE XXI. Extent of qualification by style.

Style	Will qualify style(s)
RZR10	RZR10
RZR11	RZR11, RZR10
RZR20	RZR20, RZR10
RZR21	RZR21, RZR20, RZR11, RZR10
RZR30	RZR30
RZR31	RZR31, RZR30
RZR40	RZR40
RZR41	RZR41, RZR40
RZR50	RZR40, RZR50
RZR51	RZR51, RZR50, RZR41, RZR40
RZR60	RZR60, RZR50, RZR40
RZR61	RZR61, RZR60, RZR51, RZR50, RZR41, RZR40
RZR70	RZR70
RZR71	RZR71, RZR70
RZR80	RZR80, RZR70
RZR81	RZR81, RZR80, RZR71, RZR70
RZR90	RZR90, RZR80, RZR70
RZR91	RZR91, RZR90, RZR81, RZR80, RZR71, RZR70
RZR95	RZR30
RZR96	RZR96, RZR95, RZR31, RZR30

"

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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